Education Data Structure

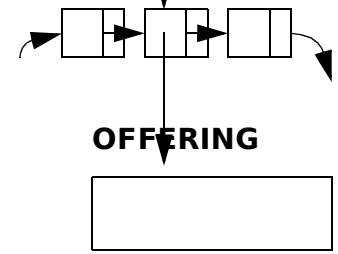
EDUCATION (1)

```
/* Usage */
EDUCATION *education_spreadsheet_fetch(
        season_name,
        year,
        spreadsheet_filename,
        date_label );

/* Input */
SEMESTER *semester =
        semester_fetch(
            season_name,
            year,
            fetch_offering_list,
            fetch_registration_list );

PAYPAL *paypal =
        paypal_spreadsheet_fetch(
            spreadsheet_filename,
            date_label );

LIST *program_list() =
        program_list(
            1 /* fetch_alias_list */ );

LIST *product_list =
        product_list();

char *spreadsheet_minimum_date =
        spreadsheet_minimum_date(
            &spreadsheet_maximum_date,
            &spreadsheet_row_count,
            spreadsheet_filename,
            date_label );
```

```
/* Process */
double education_net_revenue(
        transaction_amount,
        merchant_transaction_fee );

LIST *paypal_deposit_list =
        education_paypal_deposit_list(
            spreadsheet_filename,
            paypal->
                spreadsheet->
                spreadsheet_column_list,
            paypal->paypal_dataset,
            season_name,
            year,
            semester->offering_list(),
            this->program_list(),
            this->product_list(),
            semester->event_list() );

/* Returns paypal_dataset */
PAYPAL_DATASET *education_paypal_dataset(
        input_string,
        paypal->
            spreadsheet->
            spreadsheet_column_list(),
        paypal>paypal_dataset );

/* Cached */
LIST *education_event_list(
        char *minimum_date );
```

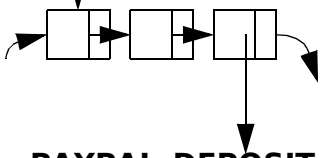
SEMESTER

```
/* Usage */
SEMESTER *semester_new(
        char *season_name,
        int year );

/* Process */
LIST *semester_offering_list(
        season_name,
        year );
```

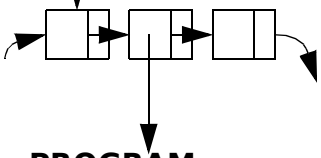


OFFERING

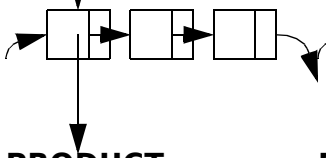
PAYPAL



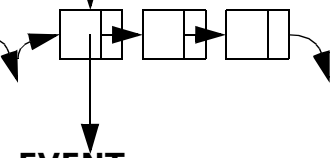
PAYPAL_DEPOSIT



PROGRAM



PRODUCT

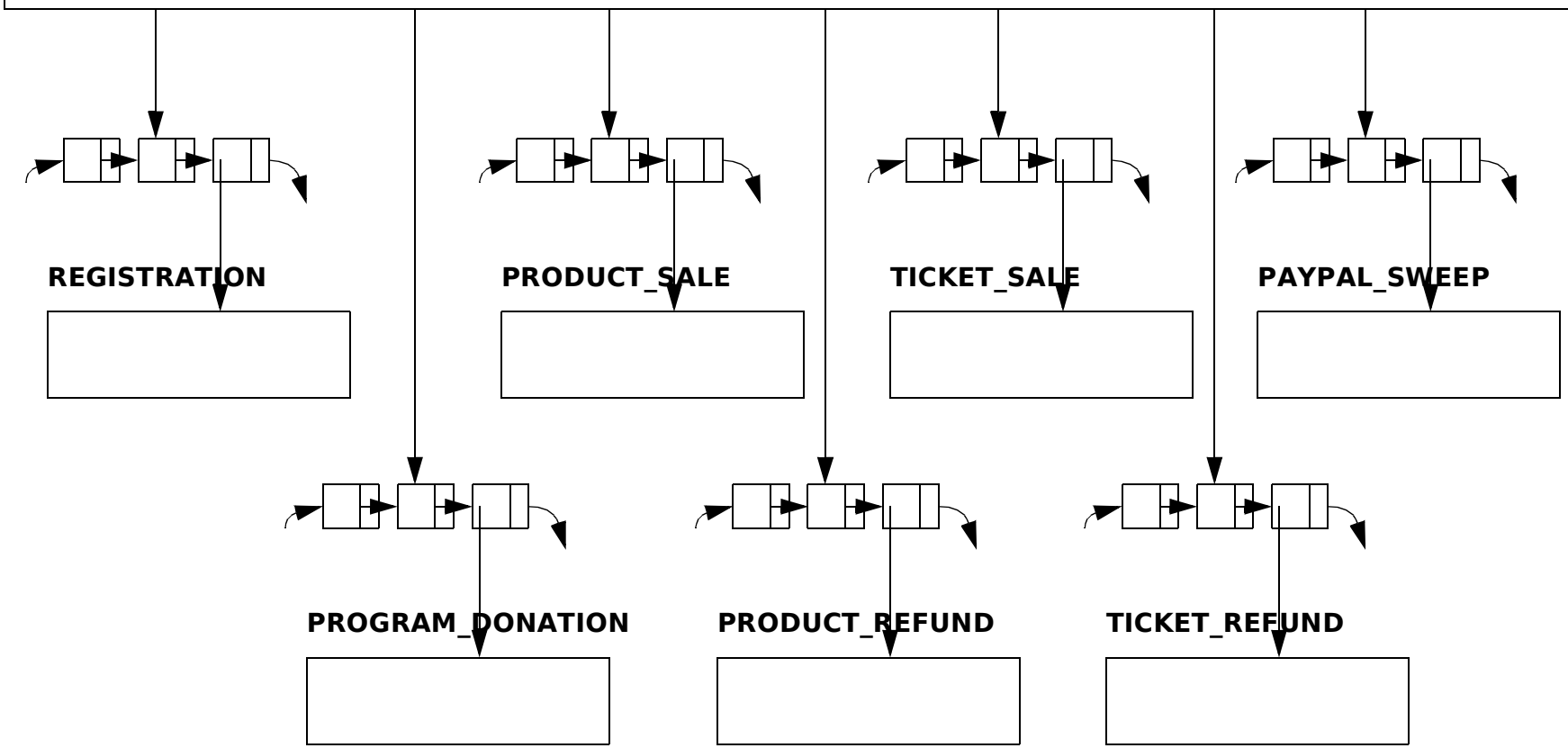


EVENT

Education Data Structure

EDUCATION (2)

```
LIST *education_existing_program_donation_list(
        spreadsheet_minimum_date() );

LIST *education_existing_tuition_payment_list(
        spreadsheet_minimum_date() );

LIST *education_existing_tuition_refund_list(
        spreadsheet_minimum_date() );

LIST *education_existing_product_sale_list(
        spreadsheet_minimum_date() );

LIST *education_existing_product_refund_list(
        spreadsheet_minimum_date() );

LIST *education_existing_ticket_sale_list(
        spreadsheet_minimum_date() );

LIST *education_existing_ticket_refund_list(
        spreadsheet_minimum_date() );

LIST *education_existing_paypal_sweep_list(
        spreadsheet_minimum_date() );

/* Returns paypal_deposit_list */
LIST *education_paypal_existing_transaction_set(
        paypal_deposit_list,
        education_existing_program_donation_list(),
        education_existing_tuition_payment_list(),
        education_existing_tuiton_refund_list(),
        education_existing_product_sale_list(),
        education_existing_product_refund_list(),
        education_existing_ticket_sale_list(),
        education_existing_ticket_refund_list(),
        education_existing_paypal_sweep_list() );
```



PAYPAL_DEPOSIT (1)

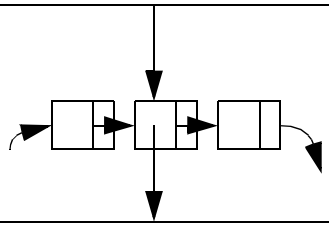
```
/* Usage */
PAYPAL_DEPOSIT *paypal_deposit_education(
        season_name,
        year,
        semester->offering_list(),
        program_list(),
        product_list(),
        semester->event_list(),
        paypal_dataset,
        row_number,
        /* Seek out existing registrations in this file */
        paypal_deposit_list() );

/* Input */
SEMESTER *semester =
        semester_new(
            season_name,
            year );
```

```
char *paypal_date_time;
double paypal_amount;
double transaction_fee;
int quantity;
char *from_email_address;
double account_balance;
char *transaction_ID;
char *invoice_number;
```

Education Data Structure

PAYPAL_DEPOSIT (2)

```
/* Process */
int row_number;

ENTITY *payor_entity =
        entity_full_name_entity(
            paypal_dataset->full_name_D );

LIST *paypal_item_list = paypal_item_list(
        paypal_item_entity_delimit(
            education_paypal_dataset()->
                item_title_P ),
        education_paypal_dataset()->
            transaction_type_E,
        education_paypal_allowed_list(
            offering_name_list()
            program_name_list(),
            program_alias_name_list(),
            product_name_list(),
            event_name_list() ) );

void paypal_deposit_set_expected_revenue(
        paypal_item_list(),
        semester_offering_list(),
        product_list(),
        education_event_list() );

LIST *registration_list =
        registration_list_paypal(
            season_name,
            year,
            payor_entity,
            paypal_date_time,
            paypal_item_list,
            semester_offering_list(),
            /* Seek out existing enrollments */
            this->paypal_deposit_list );

LIST *program_donation_list =
        program_donation_list_paypal();

LIST *product_sale_list =
        product_sale_list_paypal();

LIST *product_refund_list =
        product_refund_list_paypal();

LIST *ticket_sale_list =
        ticket_sale_list_paypal();

LIST *ticket_refund_list =
        ticket_refund_list_paypal();
```



PAYPAL_ITEM

```
/* Usage */
LIST *paypal_item_list(
        entity_delimited_item_title_P,
        transaction_type_E,
        allowed_item_list,
        event_label_list );

/* Input */
double expected_revenue;

/* Process */
char *item_data;
boolean taken;

ENTITY *benefit_entity =
        entity_full_name_entity(
            item_title_P_full_name );

double paypal_item_value(
        paypal_amount,
        this->expected_revenue,
        paypal_item_expected_revenue_total(),
        paypal_item_nonexpected_revenue_length() );

double paypal_item_fee(
        paypal_amount,
        transaction_fee,
        paypal_item_value() );

double paypal_item_gain(
        paypal_amount,
        paypal_item_expected_revenue_total(),
        paypal_item_nonexpected_revenue_length(),
        paypal_item_expected_revenue_length() );

LIST *paypal_item_steady_state_list(
        paypal_item_list,
        paypal_amount,
        transaction_fee,
        paypal_item_expected_revenue_total(),
        paypal_item_nonexpected_revenue_length(),
        paypal_item_expected_revenue_length() );
```

Education Data Structure

PAYPAL_DEPOSIT (3)

```
double tuition_payment_total(
        registration_list_paypal() );

double program_donation_total(
        program_donation_list_paypal() );

double product_sale_total(
        product_sale_list_paypal() );

double tuition_refund_total(
        registration_list_paypal() );

double product_refund_total(
        product_refund_list_paypal() );

double ticket_sale_total(
        ticket_sale_list_paypal() );

double ticket_refund_total(
        ticket_refund_list_paypal() );

PAYPAL_DEPOSIT *paypal_deposit_steady_state(
        this,
        semester_offering_list() );

void paypal_deposit_insert(
        payor_entity->full_name,
        payor_entity->street_address,
        paypal_date_time,
        row_number,
        season_name,
        year,
        paypal_amount,
        transaction_fee,
        net_revenue,
        tuition_payment_total(),
        program_donation_total(),
        ticket_sale_total(),
        product_sale_total(),
        sweep_amount(),
        tuition_refund_total(),
        ticket_refund_total(),
        product_refund_total(),
        quantity,
        from_email_address,
        account_balance,
        transaction_ID,
        invoice_number );

PAYPAL_SWEEP *paypal_sweep(
        paypal_item_list(),
        paypal_deposit->paypal_date_time );

boolean existing_transaction =
        paypal_deposit_existing_transaction(
            paypal_deposit,
            education_existing_program_donation_list(),
            education_existing_tuition_payment_list(),
            education_existing_tuition_refund_list(),
            education_existing_product_sale_list(),
            education_existing_product_refund_list(),
            education_existing_ticket_sale_list(),
            education_existing_ticket_refund_list(),
            education_existing_paypal_sweep_list() );
```

PAYPAL_SWEEP

```
/* Usage */
PAYPAL_SWEEP *paypal_sweep_fetch(
        payor_full_name,
        payor_street_address,
        paypal_date_time,
        fetch_transaction );

char *payor_full_name;
char *payor_street_address;
char *paypal_date_time;
double sweep_amount;

/* Process */
TRANSACTION *paypal_sweep_transaction(
        payor_full_name,
        payor_street_address,
        paypal_date_time,
        sweep_amount,
        entity_self_paypal_cash_account_name,
        self_cash_account() );
```

TRANSACTION

Education Data Structure

OFFERING

```
/* Usage */
OFFERING *offering_fetch(
        course_name,
        season_name,
        year,
        fetch_course,
        fetch_program );

OFFERING *offering_new(
        course_name,
        semester );

/* Input */
char *couse_name;
SEMESTER *semester;
double course_price;

ENTITY *instructor_entity =
        entity_new(
            instructor_full_name;
            instructor_street_address );

int class_capacity;
int enrollment_count;
int capacity_available;
char *revenue_account;

/* Process */
COURSE *course;
OFFERING *offering_parse(
        input,
        fetch_course,
        fetch_program )

LIST *offering_name_list(
        offering_list );

void offering_list_fetch_update(
        offering_list );

void offering_fetch_update(
        course_name,
        season_name,
        year );
```

COURSE

```
char *course_name;
char *program_name;
PROGRAM *program;

COURSE *course_fetch(
        course_name,
        fetch_program,
        fetch_alias_list );
```

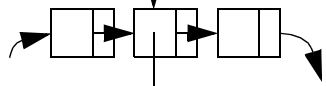
SEMESTER

ENTITY

PROGRAM

PROGRAM

```
/* Usage */
PROGRAM *program_fetch(
        char *program_name,
        boolean fetch_alias_list );

LIST *program_list_fetch(
        boolean fetch_alias_list );

/* Input */
char *program_name;
char *revenue_account;

/* Process */
LIST *program_alias_list(
        program_name );

PROGRAM *program_list_seek(
        program_name,
        program_list() );

LIST *program_name_list(
        program_list );

void program_list_fetch_update(
        program_name_list );

void program_fetch_update(
        program_name );
```



PROGRAM_ALIAS

```
char *program_name;
char *alias_name;

LIST *program_alias_fetch_list(
        program_name );

PROGRAM_ALIAS *program_alias_list_seek(
        program_alias_name,
        program_alias_list );

LIST *program_alias_name_list(
        program_list );
```

Education Data Structure

PROGRAM_DONATION

```
/* Usage */
PROGRAM_DONATION *program_donation_fetch(
        program_name,
        payor_full_name,
        payor_street_address,
        payment_date_time,
        fetch_program,
        fetch_transaction );

PROGRAM_DONATION *program_donation_new(
        program_name,
        payor_full_name,
        payor_street_address,
        payment_date_time );

/* Input */
char *program_name;
ENTITY *payor_entity;
char *payment_date_time;
double donation_amount;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
PROGRAM *program;

double net_payment_amount =
        education_net_payment_amount(
            donation_amount,
            merchant_fees_expense );

PROGRAM_DONATION *program_donation_steady_state(
        this,
        donation_amount,
        merchant_fees_expense );

PROGRAM_DONATION *program_donation_parse(
        input,
        fetch_program,
        fetch_transaction );

LIST *program_donation_list_paypal(
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        program_list() );

PROGRAM_DONATION *program_donation_paypal(
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        program );

TRANSACTION *program_donation_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        payment_date_time
            /* transaction_date_time */,
        program_name,
        donation_amount,
        merchant_fees_expense,
        education_net_payment_amount(),
        cash_account_name,
        account_fees_expense(),
        program->revenue_account );

void program_donation_update(
        education_net_payment_amount(),
        program_donation_transaction()->
            transaction_date_time,
        program_name,
        payor_entity->full_name,
        payor_entity->street_address,
        payment_date_time );

LIST *program_donation_program_name_listI(
        program_donation_list );

PROGRAM_DONATION *program_donation_list_seek(
        program_name,
        payor_full_name,
        payor_street_address,
        payment_date_time,
        program_donation_list );
```

PROGRAM

TRANSACTION

Education Data Structure

PRODUCT

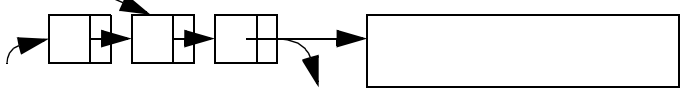
```
/* Usage */
PRODUCT *product_fetch(
        char *product_name,
        boolean fetch_program );

PRODUCT *product_new(
        char *product_name );

/* Input */
char *product_name;
char *program_name;
char *revenue_account;
double retail_price;

/* Process */
PROGRAM *program;
double sale_total;
double refund_total;

void product_list_fetch_update(
        product_name_list );

void product_fetch_update(
        product_name );

PRODUCT *product_list_seek(
        product_name,
        product_list );

LIST *product_name_list(
        product_list );
```

PROGRAM

PRODUCT_SALE

```
/* Usage */
PRODUCT_SALE *product_sale_fetch(
        product_name,
        payor_full_name,
        payor_street_address,
        sale_date_time,
        fetch_product,
        fetch_program,
        fetch_transaction );

/* Input */
char *product_name;
ENTITY *payor_entity;
char *sale_date_time;
int quantity;
double retail_price;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
PRODUCT *product;

double product_sale_extended_price(
        quantity,
        retail_price );

double net_payment_amount =
        education_net_payment_amount(
            product_sale_extended_price(),
            merchant_fees_expense );

PRODUCT_SALE *product_sale_parse(
        input,
        fetch_product,
        fetch_transaction );

PRODUCT_SALE *product_sale_steady_state(
        this,
        quantity,
        retail_price,
        merchant_fees_expense );

LIST *product_sale_list_paypal(
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        product_list() );

PRODUCT_SALE *product_sale_paypal(
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        product );

TRANSACTION *product_sale_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        sale_date_time,
        product_name,
        product->program_name,
        product_sale_extended_price(),
        merchant_fees_expense,
        education_net_payment_amount(),
        entity_self_paypal_cash_account_name(),
        account_fees_expense(),
        product->revenue_account );

void product_sale_update(
        product_sale_extended_price(),
        education_net_payment_amount(),
        product_sale_transaction()->
            transaction_date_time,
        product_name,
        payor_entity->full_name,
        payor_entity->street_address,
        sale_date_time );

LIST *product_sale_list_product_name_list(
        product_sale_list );

PRODUCT_SALE *product_sale_list_seek(
        product_name,
        payor_full_name,
        payor_street_address,
        sale_date_time,
        product_sale_list );
```

product_sale_list_paypal()

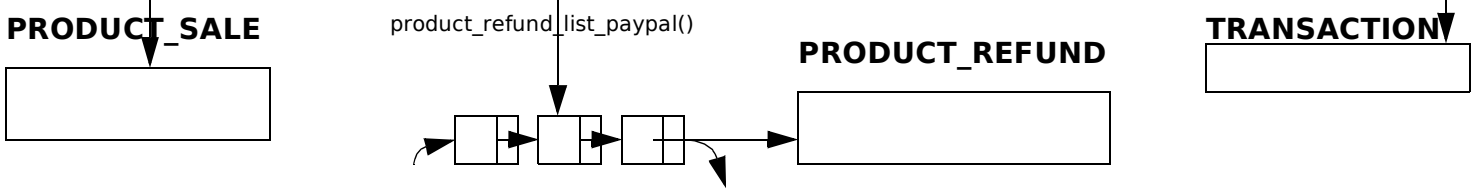


PRODUCT_SALE

TRANSACTION

PRODUCT_REFUND

```
/* Usage */
PRODUCT_REFUND *product_refund_fetch(
        product_name,
        payor_full_name,
        payor_street_address,
        sale_date_time,
        fetch_sale,
        fetch_product,
        fetch_transaction );

PRODUCT_REFUND *product_refund_new(
        product_name,
        payor_entity,
        sale_date_time );

/* Input */
char *product_name;
ENTITY *payor_entity;
char *sale_date_time;
double extended_price;
double refund_amount;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
char *refund_date_time =
        date_now19();

PRODUCT_SALE *product_sale;

double net_refund_amount =
        education_net_refund_amount(
            refund_amount,
            merchant_fees_expense );

PRODUCT_REFUND *product_refund_parse(
        input,
        fetch_sale,
        fetch_product,
        fetch_transaction );

PRODUCT_REFUND *product_refund_steady_state(
        this,
        refund_amount,
        merchant_fees_expense,
        refund_date_time );

LIST *product_refund_list_paypal(
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        product_list() );

PRODUCT_REFUND *product_refund_paypal(
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        product );

TRANSACTION *product_refund_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        refund_date_time,
        product_name,
        this->
            product_sale->
            program->
            program_name,
        refund_amount,
        merchant_fees_expense,
        education_net_payment_amount(),
        cash_account_name,
        account_fees_expense(),
        this->
            product_sale->
            product->
            revenue_account );

void product_refund_insert_pipe(
        insert_pipe,
        product_name,
        payor_full_name,
        payor_street_address,
        sale_date_time,
        refund_date_time,
        extended_price,
        refund_amount,
        merchant_fees_expense,
        net_payment_amount,
        transaction_date_time,
        paypal_date_time );

double product_refund_amount(
        refund_amount );

void product_refund_update(
        product_refund_amount(),
        date_now19() /* refund_date_time */,
        education_net_refund_amount(),
        product_refund_transaction()->
            transaction_date_time,
        product_name,
        payor_entity->full_name,
        payor_entity->street_address,
        sale_date_time );

LIST *product_refund_list_product_name_list(
        product_refund_list );

PRODUCT_REFUND *product_refund_list_seek(
        product_name,
        payor_full_name,
        payor_street_address,
        sale_date_time );
```

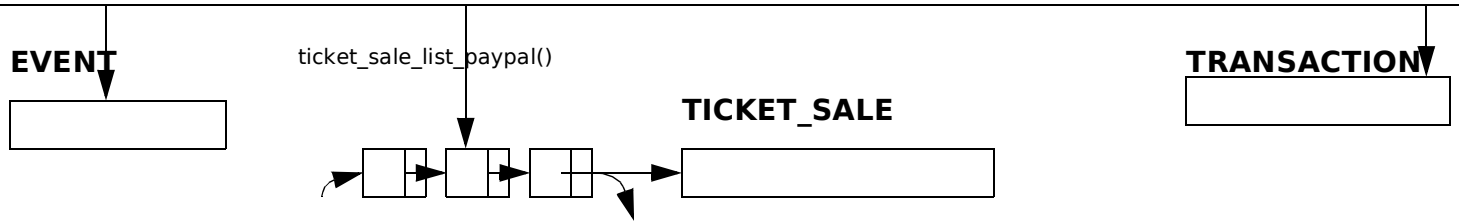


EVENT

```
/* Usage */
EVENT *event_fetch(
        char *program_name,
        char *event_date,
        char *event_time,
        boolean fetch_program,
        boolean fetch_venue );

EVENT *event_new(
        char *program_name,
        char *event_date,
        char *event_time );

LIST *event_system_list(
        event_system_string(),
        fetch_program,
        fetch_venue );

/* Input */
char *program_name;
char *event_date;
char *event_time;
char *venue_name;
char *venue_street_address;
double ticket_price;
char *revenue_account;

/* Process */
PROGRAM *program;
VENUE *venue;

char *event_primary_where(
        program_name,
        event_date,
        event_time );

char *event_system_string(
        where );

LIST *event_list_program_name_list(
        event_list );

EVENT *event_list_seek(
        program_name,
        event_date,
        event_time,
        event_list );

void event_list_fetch_update(
        event_list );

void event_fetch_update(
        program_name,
        event_date,
        event_time );

/* Paypal */
LIST *event_label_list(
        event_list );

char *event_label_long_display(
        program_name,
        event_date,
        event_time );

char *event_label_short_display(
        program_name,
        event_date,
        event_time );

EVENT *event_long_label_seek(
        event_label_long_display(),
        event_list );

EVENT *event_short_label_seek(
        event_label_short_display(),
        event_list );
```

TICKET_SALE

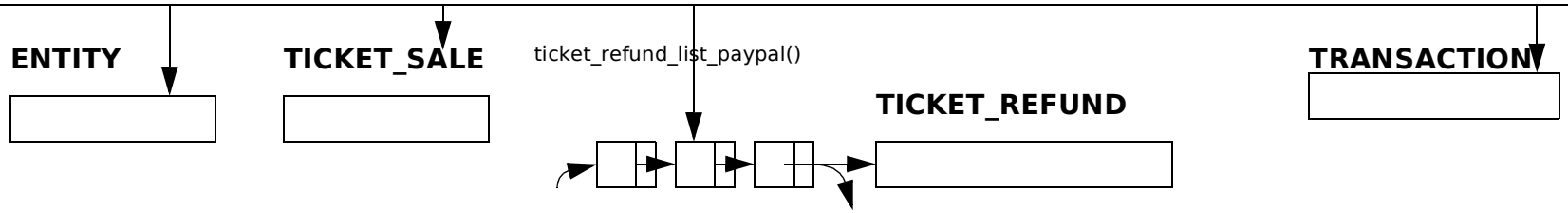
```
/* Usage */
TICKET_SALE *ticket_sale_fetch(
        program_name,
        event_date,
        event_time,
        payor_full_name,
        payor_street_address,
        fetch_event,
        fetch_transaction );

TICKET_SALE *ticket_sale_new(
        program_name,
        event_date,
        event_time,
        payor_entity );

LIST *ticket_sale_system_list(
        ticket_sale_system_string(),
        fetch_event,
        fetch_transaction );

/* Input */
char *program_name;
char *event_date;
char *event_time;
ENTITY *payor_entity;
int quantity;
double ticket_price;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
char *ticket_sale_date_time =
        date_now19();
EVENT *event;

char *ticket_sale_system_string(
        where );

double ticket_sale_extended_price(
        quantity,
        ticket_price );

double net_payment_amount =
        education_net_payment_amount(
            ticket_sale_extended_price(),
            merchant_fees_expense );

TICKET_SALE *ticket_sale_steady_state(
        this,
        quantity,
        ticket_price,
        merchant_fees_expense,
        sale_date_time );

LIST *ticket_sale_list_paypal(
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        education_event_list(),
        paypal_amount );

TICKET_SALE *ticket_sale_paypal(
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        event,
        paypal_amount,
        paypal_item_list_length );

TRANSACTION *ticket_sale_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        sale_date_time
            /* transaction_date_time */,
        event->program->program_name,
        ticket_sale_extended_price(),
        merchant_fees_expense,
        education_net_payment_amount(),
        cash_account_name(),
        account_fees_expense(),
        event->revenue_account );

void ticket_sale_update(
        date_now19() /* sale_date_time */,
        ticket_sale_extended_price(),
        education_net_payment_amount(),
        ticket_sale_transaction()->
            transaction_date_time,
        program_name,
        event_date,
        event_time,
        payor_entity->full_name,
        payor_entity->street_address,
        sale_date_time );

LIST *ticket_sale_list_event_list(
        ticket_sale_list );
```



Education Data Structure

TICKET_REFUND

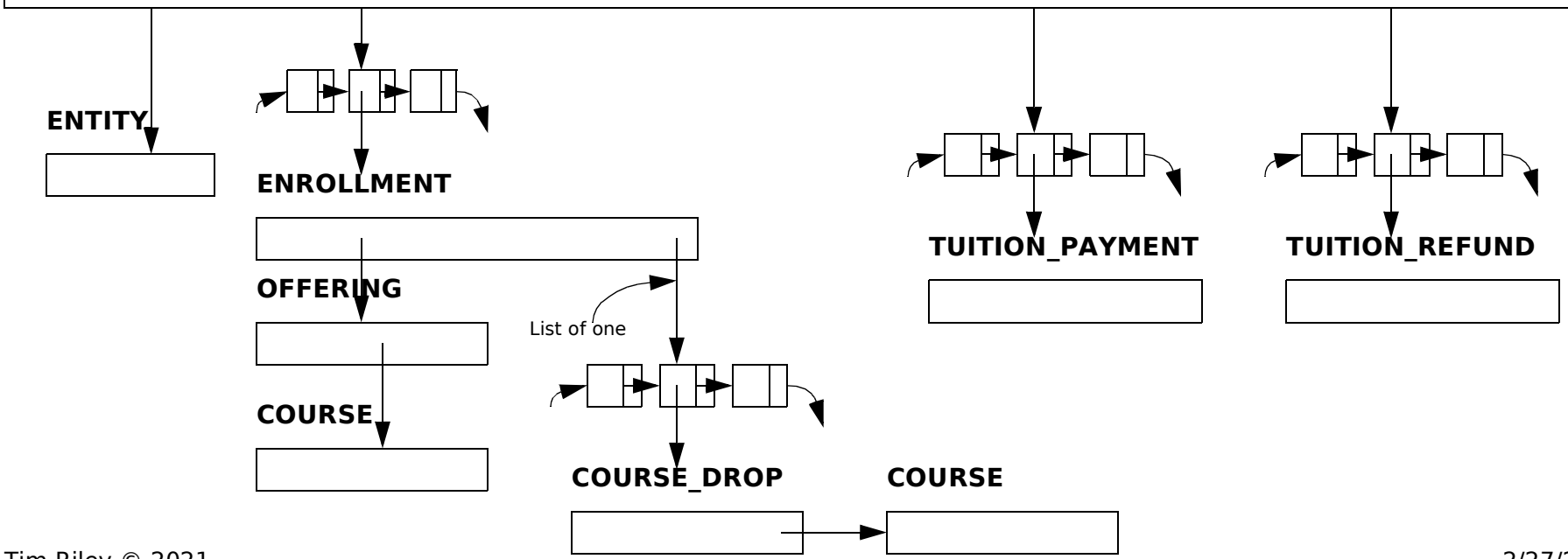
```
/* Usage */
TICKET_REFUND *ticket_refund_fetch(
        program_name,
        event_date,
        event_time,
        payor_full_name,
        payor_street_address,
        fetch_sale,
        fetch_event,
        fetch_transaction );

TICKET_REFUND *ticket_refund_new(
        program_name,
        event_date,
        event_time,
        payor_entity );

LIST *ticket_refund_system_list(
        ticket_refund_system_string(),
        fetch_sale,
        fetch_event,
        fetch_transaction );

/* Input */
char *program_name;
char *event_date;
char *event_time;
ENTITY *payor_entity;
double extended_price;
double refund_amount;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
TICKET_SALE *ticket_sale;

char *refund_date_time =
        date_now19();

double net_refund_amount =
        education_net_refund_amount(
            refund_amount,
            merchant_fees_expense );

TICKET_REFUND *ticket_refund_parse(
        input,
        fetch_sale,
        fetch_event,
        fetch_transaction );

TICKET_REFUND *ticket_refund_steady_state(
        this,
        refund_amount,
        merchant_fees_expense,
        refund_date_time );

LIST *ticket_refund_list_paypal(
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        education_event_list() );

TUITION_REFUND *ticket_refund_paypal(
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        event );

TRANSACTION *ticket_refund_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        refund_date_time
            /* transaction_date_time */,
        event_name,
        program_name,
        refund_amount,
        merchant_fees_expense,
        education_net_refund_amount(),
        cash_account_name,
        account_fees_expense(),
        this->
            ticket_sale->
            event->
            revenue_account );

void ticket_refund_update(
        date_now19() /* refund_date_time */,
        education_net_refund_amount(),
        ticket_refund_transaction()->
            transaction_date_time,
        program_name,
        event_date,
        event_time,
        payor_entity->full_name,
        payor_entity->street_address );
```



Education Data Structure

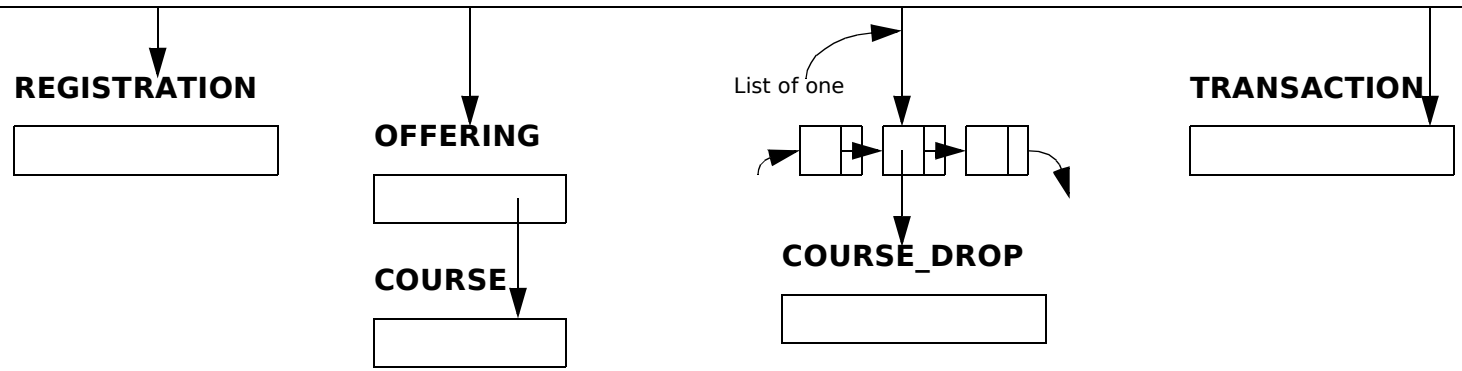
REGISTRATION

```
/* Usage */
REGISTRATION *registration_fetch(
        student_full_name,
        student_street_address,
        season_name,
        year );

REGISTRATION *registration_new(
        student_entity,
        semester );

/* Input */
ENTITY *student_entity;
SEMESTER *semester;
ENTITY *payor_entity;

/* Process */
char *registration_date_time =
        date_now19();

double tuition;
double tuition_payment_total;
double tuition_refund_total;
double invoice_amount_due;
LIST *enrollment_list;
LIST *tuition_payment_list;
LIST *tuition_refund_list;

LIST *registration_list_paypal(
        season_name,
        year,
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        semester_offering_list(),
        /* Seek out existing enrollments for refunds */
        paypal_deposit_list );

REGISTRATION *registration_paypal(
        season_name,
        year,
        paypal_item->benefit_entity
            /* student_entity */,
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        offering,
        /* Seek out existing enrollments for refunds */
        paypal_deposit_list );

void registration_update(
        date_now19() /* registration_date_time */,
        student_full_name,
        student_street_address,
        season_name,
        year );

void registration_list_fetch_update(
        LIST *registration_list );

void registration_fetch_update(
        student_full_name,
        student_street_address,
        season_name,
        year );

LIST *registration_list_course_name_list(
        registration_list );

LIST *registration_list_enrollment_list(
        registration_list );

LIST *registration_list_tuition_payment_list(
        registration_list );

LIST *registration_list_tuition_refund_list(
        registration_list );
```



ENROLLMENT

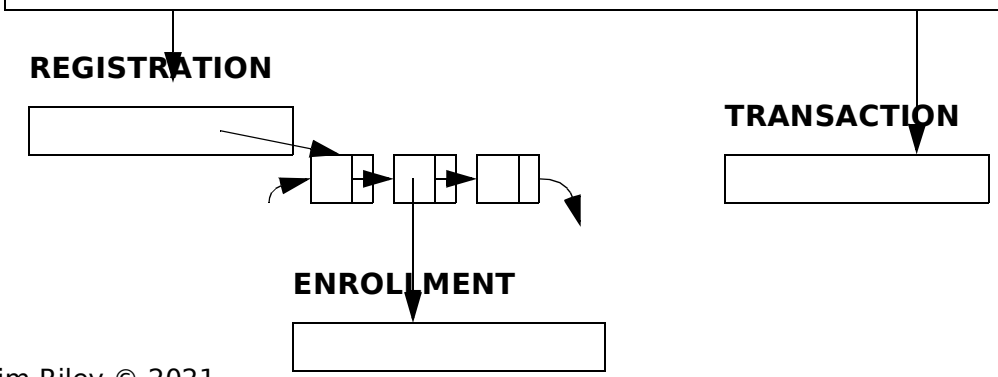
```
/* Usage */
ENROLLMENT *enrollment_fetch(
        student_full_name,
        street_address,
        course_name,
        season_name,
        year,
        fetch_offering,
        fetch_course,
        fetch_program,
        fetch_registration,
        fetch_course_drop_list,
        fetch_transaction );

ENROLLMENT *enrollment_new(
        student_entity,
        course,
        semester );

ENTITY *student_entity;
char *course_name;
SEMESTER *semester;

/* Process */
char *enrollment_date_time =
        date_now19();

COURSE *course;
REGISTRATION *registration;
OFFERING *offering;

/* List of one */
LIST *course_drop_list;

char *enrollment_primary_where(
        student_full_name,
        street_address,
        course_name,
        season_name,
        year );

TRANSACTION *enrollment_transaction(
        *seconds_to_add,
        student_full_name,
        street_address,
        enrollment_date_time,
        this->course->program_name,
        this->course->course_price,
        account_receivable(),
        offering_revenue_account(),
        account_payable(),
        liability_steady_state_entity_list() );

char *enrollment_list_first_program_name(
        enrollment_list );

ENROLLMENT *enrollment_list_seek(
        student_full_name,
        street_address,
        course_name,
        season_name,
        year,
        enrollment_list );

void enrollment_update(
        date_time19() /* enrollment_date_time */,
        payor_full_name,
        payor_street_address,
        enrollment_transaction()->
            transaction_date_time,
        student_full_name,
        student_street_address,
        course_name,
        season_name,
        year );
```



Education Data Structure

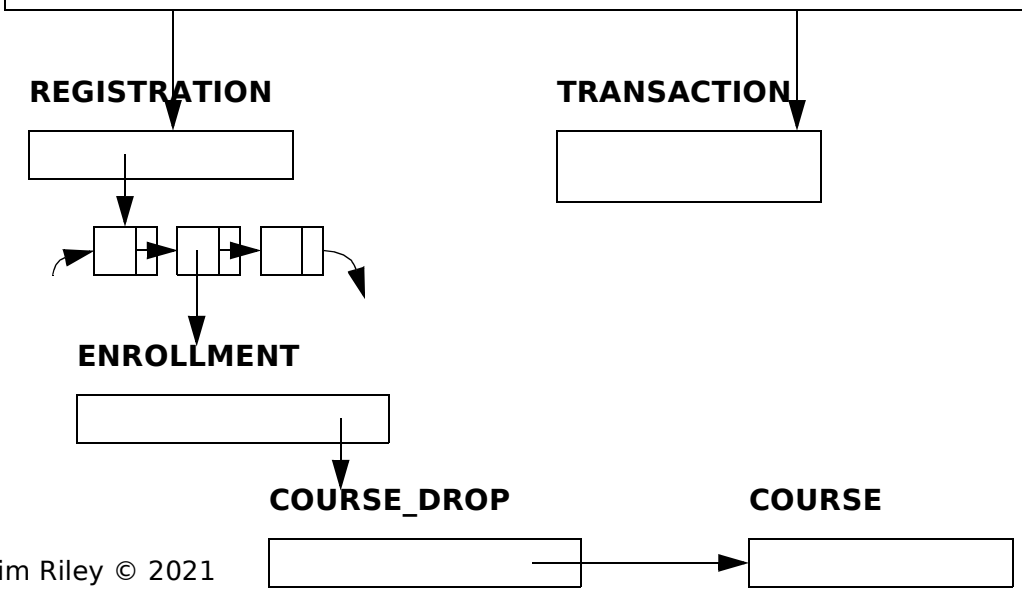
TUITION_PAYMENT

```
/* Usage */
TUITION_PAYMENT *tuition_payment_fetch(
        student_full_name,
        street_address,
        season_name,
        year,
        payor_full_name,
        payor_street_address,
        payment_date_time,
        boolean fetch_registration,
        boolean fetch_enrollment_list,
        boolean fetch_offering,
        boolean fetch_course,
        boolean fetch_program,
        boolean fetch_transaction );

TUITION_PAYMENT *tuition_payment_new(
        student_entity,
        semester,
        payor_entity,
        payment_date_time );

/* Input */
ENTITY *student_entity;
SEMESTER *semester;
ENTITY *payor_entity;
char *payment_date_time;
double payment_amount;
double merchant_fees_expense;
char *paypal_date_time;

/* Process */
REGISTRATION *registration;

double net_payment_amount =
        education_net_payment_amount(
            payment_amount,
            merchant_fees_expense );

TUITION_PAYMENT *tuition_payment_steady_state(
        this,
        payment_amount,
        merchant_fees_expense,
        semester_offering_list() );

TRANSACTION *tuition_payment_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        payment_date_time,
        offering->
            course->
            program_name,
        payment_amount,
        merchant_fees_expense,
        education_net_payment_amount(),
        account_cash(),
        account_receivable(),
        account_fees_expense() );

void tuition_payment_update(
        education_net_payment_amount(),
        tuition_payment_transaction()->
            transaction_date_time,
        student_entity->full_name,
        student_entity->street_address,
        season_name,
        year,
        payor_entity->full_name,
        payor_entity->street_address,
        payment_date_time );
```



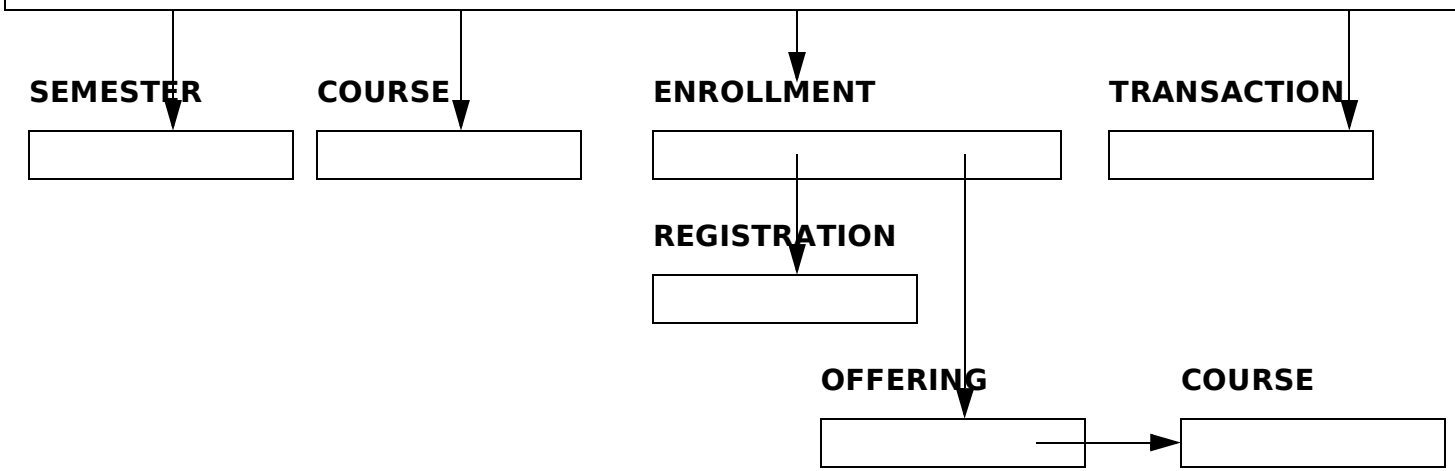
TUITION_REFUND

```
/* Usage */
TUITION_REFUND *tuition_refund_fetch(
        student_full_name,
        street_address,
        season_name,
        year,
        payor_full_name,
        payor_street_address,
        payment_date_time,
        refund_date_time,
        boolean fetch_registration,
        boolean fetch_enrollment_list,
        boolean fetch_offering,
        boolean fetch_course,
        boolean fetch_program );

TUITION_REFUND *tuition_refund_new(
        student_entity,
        semester,
        payor_entity,
        refund_date_time );

/* Input */
ENTITY *student_entity;
SEMESTER *semester;
ENTITY *payor_entity;
char *refund_date_time;
double refund_amount;
double merchange_fees_expense;
char *paypal_date_time;

/* Process */
REGISTRATION *registration;

double net_refund_amount =
        education_net_refund_amount(
            refund_amount,
            merchant_fees_expense );

TUITION_REFUND *tuition_refund_steady_state(
        this,
        refund_amount,
        merchant_fees_expense );

TRANSACTION *tuition_refund_transaction(
        *seconds_to_add,
        payor_entity->full_name,
        payor_entity->street_address,
        refund_date_time,
        offering->course->program_name,
        refund_amount,
        merchant_fees_expense,
        account_cash(),
        account_receivable(),
        account_fees_expense() );
```



COURSE_DROP

```
/* Usage */
COURSE_DROP *course_drop_fetch(
        student_full_name,
        street_address,
        course_name,
        season_name,
        year,
        boolean fetch_enrollment,
        boolean fetch_offering,
        boolean fetch_course,
        boolean fetch_registration,
        boolean fetch_transaction );

COURSE_DROP *course_drop_new(
        ENTITY *student_entity,
        char *course_name,
        SEMESTER *semester,
        char *course_drop_date_time );

/* Input */
ENTITY *student_entity;
char *course_name;
SEMESTER *semester;
char *course_drop_date_time;
boolean refund_due;
```

```
/* Process */
COURSE *course;
ENROLLMENT *enrollment;

TRANSACTION *course_drop_transaction(
        *seconds_to_add,
        student_full_name,
        street_address,
        transaction_date_time,
        enrollment->
            offering->
            course->
            program_name,
        enrollment->offering->course_price,
        enrollment->offering->revenue_account,
        account_payable() );

LIST *course_drop_list_paypal(
        season_name,
        year,
        paypal_deposit->payor_entity,
        paypal_deposit->paypal_date_time,
        paypal_item_list(),
        semester_offering_list() );
```

```
COURSE_DROP *course_drop_paypal(
        season_name,
        year,
        paypal_item->benefit_entity
            /* student_entity */,
        payor_entity,
        paypal_date_time,
        paypal_item->item_value,
        paypal_item->item_fee,
        offering );

LIST *course_drop_list_enrollment_list(
        course_drop_list );

void course_drop_update(
        date_time19() /* course_drop_date_time */,
        course_drop_transaction()->
            transaction_date_time,
        student_full_name,
        student_street_address,
        course_name,
        season_name,
        year );
```